Lab: Exploring Action

Objective

Design and simulate a simple agent that demonstrates the principles of **Action**.

Prerequisites

- Basic understanding of Python or a relevant simulation tool.
- Familiarity with the formal definition of Action.

Steps

1. **Define the Environment**: Create a simple grid world or state space relevant to Embodied.
2. **Define the Agent**: Specify the agent's generative model, focusing on Action.
3. **Simulation**: Run the agent for 100 timesteps.
4. **Perturbation**: Introduce a specific challenge to Action (e.g., increased noise, occlusion).
5. **Analysis**: Plot the Free Energy over time. Does the agent successfully adapt?

Discussion Requirements

- Attach your code or simulation logs.
- Explain the specific mechanism used to implement Action.